LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center. various forces e.g. lift, drag and moments
5. Specific Objectives	11.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Analysis of pressure distribution graphs for different angles of attack; identification of CP and AC locations; safety: No physical handling of live instruments; supervision: Instructor monitors graph interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aerofoil section?
(b) Define the angle of attack in simple terms.
(c) List the three main aerodynamic forces acting on a wing.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Pressure distribution at zero angle of attack • Effect of increasing angle of attack on pressure • Identification of stagnation and separation points • Calculation of resultant lift and drag vectors • Concept of moment generation about leading edge	L & Ds	PPT, IB	15 min
(b)	• Definition and movement of center of pressure • Definition and fixed nature of aerodynamic center • Comparison of CP and AC locations • Impact of angle of attack on CP position • Significance of AC in stability analysis	L & Ds	PPT, PDF	20 min
(c)	• Relationship between lift, drag, and moment • Graphical determination of resultant forces • Effect of stall on pressure distribution • Summary of CP and AC characteristics • Application to helicopter rotor performance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why the center of pressure moves with angle of attack.
(b) Why is the aerodynamic center preferred for stability calculations?
(c) Describe how pressure distribution changes near the stall angle.

12. Home Assignment and Follow Up	Prepare a brief note comparing the behavior of the center of pressure and aerodynamic center, referencing the concepts covered in LP 13 regarding vortex theory and lift generation.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	NACA series, Vortex theory, Magnus effect, Kutta Joukowski law, Span wise flow, down Wash wing tip and trailing edge vortices. Methods for preventing span-wise flow.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Instructor, P No.	Senior Instructor, Senior Instructor, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min